Applied Econometrics: Session 2

Making Regression Make Sense: Selection on Observables

PhD Intensive Course

Lecture 2

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Why This Lecture Matters

Lecture 1 gave the causal language: potential outcomes, selection bias, and random assignment.

Lecture 2 asks a harder question:

Core question

When we run a regression in observational data, what does that coefficient mechanically estimate, and when can we read it as causal?

- Regression is always a statistical approximation device.
- Causality is an extra layer, not a free gift from software.

Regression as a Summary of the CEF

- Start from the object we want to summarize: the conditional expectation function.
- Then define regression as a disciplined linear approximation to that object.
- Only after that do we ask whether the CEF itself is causal.

Why Regression Shows Up Everywhere

- In experiments, regression with a treatment dummy reproduces treatment-control mean differences.
- In observational work, regression is a control strategy for observed confounders.
- In descriptive work, regression is a compact summary of how averages vary with covariates.

Core point

Before arguing about causal interpretation, we should understand what regression always does mechanically.

The Conditional Expectation Function

For a scalar outcome Y_i and covariates X_i, define

$$m(x) \equiv \mathbb{E}[Y_i \mid X_i = x].$$

- The CEF is the average value of Y_i for units with the same covariates.
- If X_i is binary, the CEF is just two cell means.
- If X_i is continuous, the CEF is a function, often nonlinear.
- The CEF is about averages, not automatically about causality.

Example

$m(s) = \mathbb{E}[\log w_i \mid S_i = s]$ is the average log wage of workers with s years of schooling.

Law of Iterated Expectations

$$\mathbb{E}[Y_i] = \mathbb{E}\{\mathbb{E}[Y_i \mid X_i]\}.$$

- First average within covariate groups.
- Then average those group means using the distribution of X_i.
- This identity is the bridge between cell-specific effects and population averages.

Intuition

An unconditional average is just a weighted average of conditional averages.

Appendix details

CEF Decomposition Property

$$Y_i = \mathbb{E}[Y_i \mid X_i] + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid X_i] = 0.$$

- The first term is the part explained by X_i.
- The residual is mean-independent of X_i.
- Therefore the residual is uncorrelated with any function of X_i.

Why this matters

The CEF is not just descriptive; it is the unique decomposition of outcomes into an explained part and an orthogonal leftover.

Appendix details

The CEF Is the Best Unrestricted Predictor

$$\mathbb{E}[Y_i \mid X_i] = \underset{m(X_i)}{\arg\min}\, \mathbb{E}\left[(Y_i - m(X_i))^2\right].$$

- If you may use any function of X_i, the CEF minimizes mean squared prediction error.
- This is an MMSE statement, not a causal statement.
- The same logic explains why averages are central objects in econometrics.

Takeaway

The CEF is the best possible predictor given the information in X_i, but it may still be nonlinear and causally meaningless.

Population Regression Is a Linear Approximation Problem

Define the population regression coefficient vector by

$$\beta = \arg\min_{b} \mathbb{E}\left[\left(Y_i - X_i' b\right)^2\right].$$

First-order conditions imply

$$\beta = \mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i Y_i].$$

If $e_i \equiv Y_i - X_i'\beta$, then

$$\mathbb{E}[X_i e_i] = 0.$$

- The regression residual is orthogonal to the regressors by construction.
- This orthogonality does not by itself imply causality.

Regression Anatomy

In the simple regression of Y_i on a constant and x_i,

$$\beta_1 = \frac{\mathrm{Cov}(Y_i, x_i)}{\mathrm{Var}(x_i)}.$$

In a multivariate regression, the coefficient on x_{ki} is

$$\beta_k = \frac{\mathrm{Cov}(Y_i, \tilde{x}_{ki})}{\mathrm{Var}(\tilde{x}_{ki})},$$

where $\tilde{x}_{ki}$ is the residual from regressing x_{ki} on all other covariates.

- Every multiple-regression coefficient is a bivariate slope after partialling out other regressors.
- This is the algebraic core of Frisch-Waugh-Lovell intuition.

Appendix details

Three Reasons Regression Is Useful

1. If the CEF is linear, regression recovers it exactly.
2. Even if the CEF is nonlinear, regression is the best linear predictor of Y_i.
3. Most importantly, regression is the best linear approximation to the CEF itself.

$$\beta = \arg\min_b \mathbb{E}\left[\left(\mathbb{E}[Y_i \mid X_i] - X_i' b\right)^2\right].$$

Preferred interpretation

Regression is best thought of as approximating the CEF, not as solving a philosophical problem about individual prediction.

Appendix details

Saturated Models and Interactions

- If all covariates are discrete and we include a dummy for every covariate cell, regression is saturated.
- Saturated regression reproduces the cell means exactly.
- Main effects without interactions impose additivity across covariates.
- Interactions matter whenever the effect of one covariate depends on another.

Student warning

"Controlling for X" is not one thing. Controlling flexibly with interactions is very different from forcing a linear or additive specification.

Population First, Sample Second

- The population coefficient β is the object of interest.
- The sample estimator is $\hat{\beta} = (X'X)^{-1}X'Y$.
- Under random sampling and standard regularity conditions, $\hat{\beta}$ converges to β.
- Heteroskedasticity affects precision formulas, not the meaning of β.

Course sequencing

We will discuss inference in more detail later; today the priority is what the estimand means.

When Does Regression Become Causal?

- Regression always approximates a CEF.
- But only a causal CEF can justify causal language for the coefficient.
- The key assumption is conditional independence, also called selection on observables.

From Predictive CEFs to Causal CEFs

- A predictive CEF summarizes average outcomes across observed groups.
- A causal CEF summarizes average potential outcomes for a fixed reference population.
- To move from one to the other, we need a design or assumption that removes selection bias.

Translation rule

Regression inherits its interpretation from the CEF it approximates. If the CEF is causal, regression is causal. If the CEF is not causal, regression is not causal.

Selection Bias Returns in Observational Data

Let C_i denote college attendance. Then

$$\mathbb{E}[Y_i \mid C_i = 1] - \mathbb{E}[Y_i \mid C_i = 0] = \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid C_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_{0i} \mid C_i = 1] - \mathbb{E}[Y_{0i} \mid C_i = 0]}_{\text{selection bias}}.$$

- The observed difference combines a causal term and a non-causal baseline term.
- In schooling applications, the bias is often thought to be positive because college-goers would earn more even without college.

Appendix details

The Conditional Independence Assumption

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp C_i \mid X_i.$$

- Conditional on observed covariates X_i, treatment is as good as random.
- Equivalently, once we hold X_i fixed, treated and untreated units are comparable in potential outcomes.
- This is why the assumption is called selection on observables.

Important

CIA is not testable from the observed outcome alone. It is a research-design claim about the treatment assignment process.

What CIA Buys Us

Under CIA,

$$\mathbb{E}[Y_i \mid X_i, C_i = 1] - \mathbb{E}[Y_i \mid X_i, C_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i].$$

- Within an X_i-cell, treated and untreated units can be compared causally.
- Selection bias vanishes after conditioning on X_i.
- The price is strong: every confounder that matters must be observed and correctly controlled.

Multi-Valued Treatment and Potential Outcome Schedules

When treatment intensity is S_i rather than a binary indicator, write

$$Y_{si} \equiv f_i(s).$$

CIA becomes

$$Y_{si} \perp\!\!\!\perp S_i \mid X_i \quad \text{for all } s.$$

Then

$$\mathbb{E}[Y_i \mid X_i, S_i = s] - \mathbb{E}[Y_i \mid X_i, S_i = s-1] = \mathbb{E}[f_i(s) - f_i(s-1) \mid X_i].$$

- This is the logic behind causal interpretations of schooling regressions.
- The treatment effect may vary with both s and X_i.

From X-Specific Effects to Population Effects

The law of iterated expectations turns cell-specific causal effects into summary effects:

$$\mathrm{ATE} = \mathbb{E}\left\{\mathbb{E}[Y_i \mid X_i, S_i = s] - \mathbb{E}[Y_i \mid X_i, S_i = s-1]\right\}.$$

The effect on those actually treated at s uses different weights:

$$\mathbb{E}\left\{\mathbb{E}[Y_i \mid X_i, S_i = s] - \mathbb{E}[Y_i \mid X_i, S_i = s-1] \mid S_i = s\right\}.$$

- Same cell-specific effects.
- Different target populations.
- Different weights over X_i.

A Linear Causal Model That Leads to Regression

Suppose potential outcomes satisfy

$$f_i(s) = \alpha + \rho s + \eta_i.$$

If

$$\eta_i = X_i'\gamma + v_i \qquad \text{and CIA holds given } X_i,$$

then the observed outcome obeys

$$Y_i = \alpha + \rho S_i + X_i'\gamma + v_i,$$

with v_i orthogonal to S_i and X_i.

Interpretation

In this special case, the regression coefficient ρ is the causal return to one additional unit of treatment.

Appendix details

What Counts as a Good Control?

Good controls are variables that:

- are determined before treatment,
- help explain both treatment assignment and outcomes,
- make the CIA more plausible,
- do not themselves lie on the causal path from treatment to outcome.

Rule of thumb

Good controls are confounders or proxies for confounders. They are not consequences of treatment.

The Omitted Variables Bias Formula

Suppose the long regression is

$$Y_i = \alpha + \rho S_i + A_i'\gamma + \varepsilon_i,$$

where A_i denotes omitted ability or family-background variables. Then the short-regression slope satisfies

$$\frac{\mathrm{Cov}(Y_i, S_i)}{\mathrm{Var}(S_i)} = \rho + \gamma'\delta_{AS},$$

where δ_{AS} is the vector of coefficients from regressions of A_i on S_i.

Memorize this

Short = long + effect of omitted × correlation of omitted with included.

Appendix details

Sign Intuition for OVB

- If omitted ability raises wages, then elements of γ are positive.
- If ability is positively correlated with schooling, then δ_{AS} is positive.
- The short schooling regression overstates the causal return.

But do not overlearn the sign

OVB is not always upward. If omitted factors are negatively correlated with schooling, bias can go the other way.

Illustrative Schooling Example

Specification	Schooling Coefficient
No controls	0.132
+ Age dummies	0.131
+ Family background and demographics	0.114
+ AFQT score	0.087
+ Occupation dummies	0.066

- The drop from 0.132 to 0.087 is consistent with positive omitted-ability bias.
- The final drop after adding occupation dummies is harder to interpret.

Why harder?

Occupation is likely affected by schooling, so it is not a clean pre-treatment control.

Bad Controls

A bad control is a variable that is itself an outcome of the treatment.

- Example: controlling for occupation in a regression of wages on college.
- College affects occupation.
- Conditioning on occupation therefore changes the composition of the comparison groups.

$$\mathbb{E}[Y_i \mid W_i = 1, C_i = 1] - \mathbb{E}[Y_i \mid W_i = 1, C_i = 0]$$

is generally not a causal effect.

Verbal summary

We stop comparing like with like because the treatment changes who enters the conditioned-on group.

Appendix details

Proxy Controls Can Also Go Bad

- Suppose you cannot observe latent ability directly.
- You may be tempted to control for a post-treatment proxy, such as later occupation or post-schooling test scores.
- That proxy may partly absorb the very channel through which treatment works.

Safe question

Was this variable already fixed before treatment was chosen?

Unsafe question

Is this variable merely correlated with the outcome?

Checklist for Choosing Controls

Before adding a variable X to a causal regression, ask:

1. Was X determined before treatment?
2. Could treatment itself change X?
3. Does conditioning on X make assignment more plausibly as-good-as-random?
4. Do we still have overlap between treated and control units after conditioning?

Student habit to build

Do not say "I controlled for everything available." Say exactly why each control belongs in the design.

Heterogeneity, Matching, and Propensity Scores

- Constant treatment effects and linear CEFs are convenient but not necessary.
- Under CIA, matching and regression solve the same comparability problem.
- The main differences are how they average cell-specific effects and how much modeling they impose.

Why Heterogeneity Changes the Discussion

- People may respond differently to the same treatment.
- The causal CEF may be nonlinear even if the average effect is well-defined.
- Regression then becomes a weighted average summary of many local comparisons.

Key insight

Instead of abandoning regression, reinterpret it as a computationally attractive matching estimator.

Matching the Effect of Treatment on the Treated

With binary treatment D_i, define

$$\delta_X \equiv \mathbb{E}[Y_i \mid X_i, D_i = 1] - \mathbb{E}[Y_i \mid X_i, D_i = 0].$$

Under CIA,

$$\mathrm{TOT} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] = \mathbb{E}[\delta_X \mid D_i = 1].$$

If X_i is discrete,

$$\mathrm{TOT} = \sum_x \delta_x \mathbb{P}(X_i = x \mid D_i = 1).$$

- Matching constructs δ_x cell by cell.
- Then it averages those cell effects using treated-group weights.

Appendix details

ATT and ATE Differ by Weights

Under CIA,

$$\mathrm{ATE} = \mathbb{E}[\delta_X] = \sum_x \delta_x \mathbb{P}(X_i = x),$$

while

$$\mathrm{TOT} = \sum_x \delta_x \mathbb{P}(X_i = x \mid D_i = 1).$$

- Same covariate-specific treatment effects.
- Different weighting schemes.
- Therefore ATE and TOT need not be numerically close.

Interpretation

ATE answers "what would treatment do for the population?" while TOT answers "what did treatment do for those actually treated?"

Saturated Regression as a Matching Device

Consider the regression

$$Y_i = \sum_x d_{ix}\beta_x + \delta_R D_i + \varepsilon_i,$$

where d_{ix} is a dummy for the cell $X_i = x$.

- This regression saturates the model in X_i.
- It uses the same CIA logic as exact covariate matching.
- The coefficient δ_R is not arbitrary; it is a weighted average of the same cell-specific effects δ_x.

Appendix details

Regression Is Variance-Weighted Matching

A useful result is that

$$\delta_R = \frac{\mathbb{E}[\sigma_D^2(X_i)\delta_X]}{\mathbb{E}[\sigma_D^2(X_i)]},$$

where

$$\sigma_D^2(X_i) = \mathrm{Var}(D_i \mid X_i) = p(X_i)(1 - p(X_i)).$$

- Regression puts more weight on cells where treatment status varies a lot.
- Those are cells with both treated and control units, especially when treatment probability is near one half.
- Matching for TOT instead emphasizes cells where treated units are common.

Appendix details

Why Different Weights Matter

- If treatment effects are constant across covariate cells, all reasonable averages coincide.
- If treatment effects vary with X_i, weighting matters.
- In the military-service application, those most likely to serve seem to benefit least, so TOT matching estimates are smaller than regression estimates.

Exam takeaway

"Regression and matching disagree" does not automatically mean one is wrong. They may be targeting different weighted averages.

Common Support

- For causal comparisons at covariate value x, we need both treated and untreated observations there.
- Formally, for binary treatment we want

$$0 < \mathbb{P}(D_i = 1 \mid X_i = x) < 1.$$

- If a cell contains only treated or only untreated units, δ_x is undefined.

Good news

In theory, both matching and saturated regression put zero weight on unsupported cells.

Bad news

In practice, unsaturated parametric regressions can extrapolate into unsupported regions.

The Propensity Score Theorem

Define the propensity score

$$p(X_i) \equiv \mathbb{E}[D_i \mid X_i] = \mathbb{P}(D_i = 1 \mid X_i).$$

Rosenbaum and Rubin's theorem says:

Propensity-score theorem

If $Y_{0i}, Y_{1i} \perp\!\!\!\perp D_i \mid X_i$, then $Y_{0i}, Y_{1i} \perp\!\!\!\perp D_i \mid p(X_i)$.

- A high-dimensional covariate vector can be replaced by a scalar balancing score.
- This is dimension reduction, not magic.

Appendix details

How the Propensity Score Is Used

Under CIA,

$$\mathrm{TOT} = \mathbb{E}\left\{\mathbb{E}[Y_i \mid p(X_i), D_i = 1] - \mathbb{E}[Y_i \mid p(X_i), D_i = 0] \mid D_i = 1\right\}.$$

Practical implementations:

- match treated and control units with similar estimated scores,
- stratify observations into score bins,
- weight observations by inverse probabilities,
- screen observations to impose common support before regression.

Inverse-Probability Weighting Identities

Under CIA,

$$\mathrm{ATE} = \mathbb{E}\left[\frac{Y_i D_i}{p(X_i)} - \frac{Y_i(1-D_i)}{1-p(X_i)}\right].$$

Equivalently,

$$\mathrm{ATE} = \mathbb{E}\left[\frac{(D_i - p(X_i))Y_i}{p(X_i)(1-p(X_i))}\right].$$

For the treated effect,

$$\mathrm{TOT} = \mathbb{E}\left[\frac{(D_i - p(X_i))Y_i}{(1-p(X_i))\mathbb{P}(D_i = 1)}\right].$$

Appendix details

Regression, Matching, and IPW Are One Family

A unified way to write these estimands is:

$$\mathbb{E}\left\{g(X_i)\left[\frac{Y_i D_i}{p(X_i)} - \frac{Y_i(1-D_i)}{1-p(X_i)}\right]\right\}.$$

- $g(X_i) = 1$ gives the ATE.
- $g(X_i) = p(X_i)/\mathbb{P}(D_i = 1)$ gives the treated effect.
- A regression-type choice of $g(X_i)$ reproduces the regression estimand.

Bottom line

These methods differ mostly in how they weight the same underlying cell-level comparisons.

What to Conclude About Propensity Scores

- Propensity scores are attractive when treatment assignment is easier to model than outcomes.
- They help reduce dimensionality and organize overlap diagnostics.
- But they are not automatically more causal or more robust than regression.
- Regression with flexible controls often performs very well.

Practical recommendation

Start with regression. Use propensity-score tools as complements, especially for overlap screening and dimension reduction.

NSW Lesson: Design Beats Method Branding

- Lalonde showed naive observational comparisons can fail badly.
- Dehejia and Wahba showed better overlap and better covariate control can move observational estimates toward the experimental benchmark.
- Careful regression control plus support screening can be as persuasive as propensity-score matching.

Applied lesson

Do not ask "Should I use regression or propensity scores?" Ask "What makes CIA plausible here, and where is the common support?"

Practical Issues That Cause Confusion

- Sample weights
- Grouped-data regressions
- Limited dependent variables
- Conditional-on-positive comparisons

Theme

Many practical habits are defensible only if they move us closer to the population estimand we actually care about.

When Should You Weight a Regression?

- Use sampling weights when the sample is nonrandom and weights are inverse sampling probabilities.
- Then weighted least squares targets the population regression function rather than the sample-specific one.
- Use group-size weights when regressing cell means and you want to recover the microdata regression.

Rule of thumb

Weight when weighting makes your sample look more like the population target.

Grouped Data and Weighting by Cell Size

Suppose you observe cell means $\bar{y}_x = \mathbb{E}[Y_i \mid X_i = x]$ and cell frequencies n_x.

- Regressing $\bar{y}_x$ on x with weights n_x reproduces the microdata regression.
- Unweighted regression on cell means changes the estimand because small and large cells get equal influence.

Interpretation

Weighting by cell size is not a cosmetic choice; it determines whether we are approximating the population or a fictional equally-weighted set of cells.

Appendix details

Why We Are Skeptical of Weighting for Heteroskedasticity

- Heteroskedasticity does not change the population regression coefficient.
- Poorly estimated variance weights can hurt finite-sample performance.
- If the CEF is nonlinear, weighted least squares may approximate a different object from OLS.

Practical advice

Usually keep OLS for the point estimate and use heteroskedasticity-robust standard errors for inference.

Limited Dependent Variables Do Not Kill OLS

- If treatment is randomly assigned, the coefficient on a treatment dummy still equals the difference in means.
- This remains true whether Y_i is continuous, binary, or nonnegative.
- For a binary outcome, the difference in means is also a difference in probabilities.

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i}].$$

Message

The outcome being discrete or bounded changes interpretation, not the validity of the mean comparison itself.

Why People Use Probit and Tobit Anyway

- Nonlinear models can fit bounded outcomes more naturally.
- Probit fitted values stay between zero and one.
- Tobit fitted values stay nonnegative under the latent model.

But

The raw coefficients in Probit or Tobit are not the causal effects on observed outcomes. They must be converted into finite differences or average derivatives.

Marginal Effects Are the Right Comparison Object

For a nonlinear model with covariates, compare OLS to

$$\mathbb{E}\left\{\mathbb{E}[Y_i \mid X_i, D_i = 1] - \mathbb{E}[Y_i \mid X_i, D_i = 0]\right\},$$

not to latent-index coefficients.

- OLS estimates an average effect on the observed outcome.
- Probit and Tobit coefficients live on latent scales.
- Once converted to marginal effects, nonlinear and OLS estimates are often quite similar in practice.

Conditional-on-Positive Effects Are Another Bad-Control Problem

For a nonnegative outcome,

$$\mathbb{E}[Y_i \mid D_i] = \mathbb{E}[Y_i \mid Y_i > 0, D_i]\mathbb{P}(Y_i > 0 \mid D_i).$$

Some researchers then compare

$$\mathbb{E}[Y_i \mid Y_i > 0, D_i = 1] - \mathbb{E}[Y_i \mid Y_i > 0, D_i = 0].$$

- This conditions on an outcome that treatment may itself change.
- So it is a bad-control comparison, even in a randomized experiment.

Appendix details

Why COP Effects Are Not Causal

- Treatment changes who is in the $Y_i > 0$ group.
- Therefore treated positives and control positives are different populations.
- Even if the true treatment effect were zero, the conditional-on-positive comparison could still be nonzero.

Safe object

The average effect on the observed outcome.

Unsafe object

The effect among those with positive observed outcomes after treatment assignment has already changed that set.

When Tobit-Type Models Do Make Sense

- Tobit is most defensible under true censoring.
- Example: earnings are top-coded in a survey, but the latent uncensored earnings are the object of interest.
- It is much less convincing when zeros are genuine outcomes, such as zero medical spending or zero hours worked.

Practical position

If the latent variable has no meaningful empirical interpretation, do not hide behind it.

Practical Modeling Rules for Students

1. Start with the average effect on the observed outcome.
2. Use flexible pre-treatment controls and robust standard errors.
3. Check overlap before trusting regression or matching.
4. Treat Probit/Tobit as optional curve-fitting devices, not automatic upgrades.
5. Never condition on variables created by treatment unless you have a very explicit structural reason.

Integrated Summary of Chapter 3

- The CEF is the fundamental population object; regression is its best linear approximation.
- A regression becomes causal only when the underlying CEF is causal, typically under CIA plus good controls.
- OVB explains mechanically why coefficients move when controls are added or omitted.
- Matching, propensity-score weighting, and regression all average cell-specific causal effects using different weights.
- Limited dependent variables change presentation and modeling choices, but not the centrality of average effects.

What Students Should Be Able to Do After This Lecture

- Define the CEF and explain why regression approximates it.
- State the conditional independence assumption in words and symbols.
- Use the omitted variables bias formula to reason about coefficient movements.
- Explain why occupation can be a bad control in a schooling regression.
- Distinguish ATE, TOT, and regression-weighted effects.
- Explain what the propensity score does and does not do.
- Defend when to use weights, and when not to.
- Explain why conditional-on-positive comparisons are not generally causal.

Appendix Map

- A1: Law of iterated expectations and CEF decomposition
- A2: Why regression approximates the CEF
- A3: Regression anatomy
- A4: CIA and the linear causal regression
- A5: Omitted variables bias
- A6: Bad control decomposition
- A7: Matching formulas for TOT and ATE
- A8: Regression as variance-weighted matching
- A9: Propensity-score theorem
- A10: Inverse-probability weighting identities
- A11: Grouped-data weighting
- A12: Conditional-on-positive decomposition

Appendix A1: LIE and CEF Decomposition

Start from the law of iterated expectations:

$$\mathbb{E}[Y_i] = \mathbb{E}\{\mathbb{E}[Y_i \mid X_i]\}.$$

Define

$$\varepsilon_i \equiv Y_i - \mathbb{E}[Y_i \mid X_i].$$

Then

$$\mathbb{E}[\varepsilon_i \mid X_i] = \mathbb{E}[Y_i \mid X_i] - \mathbb{E}[Y_i \mid X_i] = 0.$$

Therefore

$$Y_i = \mathbb{E}[Y_i \mid X_i] + \varepsilon_i \quad \text{with} \quad \mathbb{E}[\varepsilon_i \mid X_i] = 0.$$

Any function $h(X_i)$ is also uncorrelated with ε_i because

$$\mathbb{E}[h(X_i)\varepsilon_i] = \mathbb{E}\{h(X_i)\mathbb{E}[\varepsilon_i \mid X_i]\} = 0.$$

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Appendix A2: Why Regression Approximates the CEF

Write

$$\left(Y_i - X_i'b\right)^2 = \left(Y_i - \mathbb{E}[Y_i \mid X_i]\right)^2 + \left(\mathbb{E}[Y_i \mid X_i] - X_i'b\right)^2 + 2\left(Y_i - \mathbb{E}[Y_i \mid X_i]\right)\left(\mathbb{E}[Y_i \mid X_i] - X_i'b\right).$$

Take expectations.

- The first term does not depend on b.
- The cross term has expectation zero because $Y_i - \mathbb{E}[Y_i \mid X_i]$ is orthogonal to any function of X_i.

So minimizing $\mathbb{E}[(Y_i - X_i'b)^2]$ is equivalent to minimizing

$$\mathbb{E}\left[\left(\mathbb{E}[Y_i \mid X_i] - X_i'b\right)^2\right].$$

Hence regression is the best linear approximation to the CEF.

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Appendix A3: Regression Anatomy

Let $\tilde{x}_{ki}$ be the residual from regressing x_{ki} on all other covariates. Start from

$$Y_i = \beta_0 + \beta_k x_{ki} + \text{other terms} + e_i.$$

Take covariance with $\tilde{x}_{ki}$:

$$\mathrm{Cov}(Y_i, \tilde{x}_{ki}) = \beta_k \mathrm{Cov}(x_{ki}, \tilde{x}_{ki}),$$

because

- $\tilde{x}_{ki}$ is uncorrelated with all other regressors by construction,
- $\tilde{x}_{ki}$ is uncorrelated with the regression residual.

Since $\mathrm{Cov}(x_{ki}, \tilde{x}_{ki}) = \mathrm{Var}(\tilde{x}_{ki})$,

$$\beta_k = \frac{\mathrm{Cov}(Y_i, \tilde{x}_{ki})}{\mathrm{Var}(\tilde{x}_{ki})}.$$

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Appendix A4: CIA and the Linear Causal Regression

With binary treatment, CIA says

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp C_i \mid X_i.$$

Therefore

$$\mathbb{E}[Y_{0i} \mid X_i, C_i = 1] = \mathbb{E}[Y_{0i} \mid X_i, C_i = 0]$$

and similarly for Y_{1i}. Hence

$$\mathbb{E}[Y_i \mid X_i, C_i = 1] - \mathbb{E}[Y_i \mid X_i, C_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i].$$

For the linear causal model

$$f_i(s) = \alpha + \rho s + \eta_i, \qquad \eta_i = X_i'\gamma + v_i,$$

CIA implies

$$\mathbb{E}[f_i(s) \mid X_i, S_i] = \mathbb{E}[f_i(s) \mid X_i] = \alpha + \rho s + X_i'\gamma,$$

so the observed-outcome equation

$$Y_i = \alpha + \rho S_i + X_i'\gamma + v_i$$

has a residual orthogonal to S_i and X_i, and ρ is causal.

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Appendix A5: Omitted Variables Bias

Long regression:

$$Y_i = \alpha + \rho S_i + A_i'\gamma + \varepsilon_i.$$

The short-regression slope on S_i is

$$\frac{\mathrm{Cov}(Y_i, S_i)}{\mathrm{Var}(S_i)}.$$

Substitute the long regression into the covariance:

$$\mathrm{Cov}(Y_i, S_i) = \rho \mathrm{Var}(S_i) + \mathrm{Cov}(A_i'\gamma, S_i) + \mathrm{Cov}(\varepsilon_i, S_i).$$

The last term is zero in the long regression, so

$$\frac{\mathrm{Cov}(Y_i, S_i)}{\mathrm{Var}(S_i)} = \rho + \frac{\mathrm{Cov}(A_i'\gamma, S_i)}{\mathrm{Var}(S_i)}.$$

Writing δ_{AS} for the coefficients from regressions of A_i on S_i,

$$\frac{\mathrm{Cov}(Y_i, S_i)}{\mathrm{Var}(S_i)} = \rho + \gamma'\delta_{AS}.$$

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Appendix A6: Why Bad Controls Generate Selection Bias

Let C_i be college and W_i be white-collar status, with potential outcomes

$$Y_i = C_i Y_{1i} + (1 - C_i) Y_{0i}, \qquad W_i = C_i W_{1i} + (1 - C_i) W_{0i}.$$

Even if C_i is randomly assigned,

$$\mathbb{E}[Y_i \mid W_i = 1, C_i = 1] - \mathbb{E}[Y_i \mid W_i = 1, C_i = 0]$$

becomes

$$\mathbb{E}[Y_{1i} \mid W_{1i} = 1] - \mathbb{E}[Y_{0i} \mid W_{0i} = 1].$$

Add and subtract $\mathbb{E}[Y_{0i} \mid W_{1i} = 1]$:

$$\mathbb{E}[Y_{1i} - Y_{0i} \mid W_{1i} = 1] + \{\mathbb{E}[Y_{0i} \mid W_{1i} = 1] - \mathbb{E}[Y_{0i} \mid W_{0i} = 1]\}.$$

The second term is selection bias caused by conditioning on a post-treatment variable.

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Appendix A7: Matching Formulas

Define

$$\delta_X \equiv \mathbb{E}[Y_i \mid X_i, D_i = 1] - \mathbb{E}[Y_i \mid X_i, D_i = 0].$$

Under CIA,

$$\mathbb{E}[Y_{0i} \mid X_i, D_i = 1] = \mathbb{E}[Y_{0i} \mid X_i, D_i = 0],$$

so

$$\mathrm{TOT} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] = \mathbb{E}[\delta_X \mid D_i = 1].$$

If X_i is discrete,

$$\mathrm{TOT} = \sum_x \delta_x \mathbb{P}(X_i = x \mid D_i = 1).$$

Similarly,

$$\mathrm{ATE} = \mathbb{E}[\delta_X] = \sum_x \delta_x \mathbb{P}(X_i = x).$$

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Appendix A8: Regression as Variance-Weighted Matching

With saturated controls in X_i, the regression coefficient on D_i satisfies

$$\delta_R = \frac{\mathrm{Cov}(Y_i, \tilde{D}_i)}{\mathrm{Var}(\tilde{D}_i)} = \frac{\mathbb{E}[(D_i - \mathbb{E}[D_i \mid X_i])Y_i]}{\mathbb{E}[(D_i - \mathbb{E}[D_i \mid X_i])^2]}.$$

Replace Y_i by $\mathbb{E}[Y_i \mid D_i, X_i]$ and expand the CEF:

$$\mathbb{E}[Y_i \mid D_i, X_i] = \mathbb{E}[Y_i \mid D_i = 0, X_i] + \delta_X D_i.$$

The baseline term drops out because it is a function of X_i. What remains is

$$\delta_R = \frac{\mathbb{E}[(D_i - \mathbb{E}[D_i \mid X_i])^2 \delta_X]}{\mathbb{E}[(D_i - \mathbb{E}[D_i \mid X_i])^2]} = \frac{\mathbb{E}[\sigma_D^2(X_i)\delta_X]}{\mathbb{E}[\sigma_D^2(X_i)]}.$$

For binary treatment, $\sigma_D^2(X_i) = p(X_i)(1 - p(X_i))$.

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Appendix A9: Proof of the Propensity-Score Theorem

We want to show that $Y_{ji} \perp\!\!\!\perp D_i \mid p(X_i)$, where $j \in \{0, 1\}$. It is enough to prove that

$$\mathbb{P}(D_i = 1 \mid Y_{ji}, p(X_i))$$

does not depend on Y_{ji}. Using iterated expectations,

$$\begin{aligned}
\mathbb{P}(D_i = 1 \mid Y_{ji}, p(X_i)) &= \mathbb{E}[D_i \mid Y_{ji}, p(X_i)] \\
&= \mathbb{E}\{\mathbb{E}[D_i \mid Y_{ji}, p(X_i), X_i] \mid Y_{ji}, p(X_i)\} \\
&= \mathbb{E}\{\mathbb{E}[D_i \mid Y_{ji}, X_i] \mid Y_{ji}, p(X_i)\} \\
&= \mathbb{E}\{\mathbb{E}[D_i \mid X_i] \mid Y_{ji}, p(X_i)\},
\end{aligned}$$

where the last step uses CIA given X_i. But $\mathbb{E}[D_i \mid X_i] = p(X_i)$, so the final expression is simply $p(X_i)$, which does not depend on Y_{ji}.

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Appendix A10: Inverse-Probability Weighting Identities

Under CIA,

$$\mathbb{E}\left[\frac{Y_i D_i}{p(X_i)}\right] = \mathbb{E}\left\{\mathbb{E}\left[\frac{Y_i D_i}{p(X_i)}\middle| X_i\right]\right\} = \mathbb{E}[Y_{1i}].$$

Similarly,

$$\mathbb{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbb{E}[Y_{0i}].$$

Subtract to obtain

$$\mathrm{ATE} = \mathbb{E}\left[\frac{Y_i D_i}{p(X_i)} - \frac{Y_i(1-D_i)}{1-p(X_i)}\right].$$

Rearranging terms yields

$$\mathrm{ATE} = \mathbb{E}\left[\frac{(D_i - p(X_i))Y_i}{p(X_i)(1-p(X_i))}\right].$$

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Appendix A11: Why Group-Size Weighting Recovers the Microdata Regression

Suppose X_i takes discrete values x, and for each cell you observe

$$\bar{y}_x = \mathbb{E}[Y_i \mid X_i = x] \qquad \text{and} \qquad n_x.$$

The population regression moment is

$$\mathbb{E}[X_i Y_i] = \sum_x x\, \mathbb{E}[Y_i \mid X_i = x] \mathbb{P}(X_i = x).$$

In a sample, $\mathbb{P}(X_i = x)$ is estimated by n_x/N, so

$$\widehat{\mathbb{E}[X_i Y_i]} = \sum_x x\, \bar{y}_x \frac{n_x}{N}.$$

This is exactly the cross-moment generated by a weighted regression of $\bar{y}_x$ on x with weights n_x. Therefore weighting cell means by cell size reproduces the microdata regression target.

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Appendix A12: Conditional-on-Positive Decomposition

Consider the conditional-on-positive comparison:

$$\mathbb{E}[Y_i \mid Y_i > 0, D_i = 1] - \mathbb{E}[Y_i \mid Y_i > 0, D_i = 0].$$

In a randomized experiment this equals

$$\mathbb{E}[Y_{1i} \mid Y_{1i} > 0] - \mathbb{E}[Y_{0i} \mid Y_{0i} > 0].$$

Add and subtract $\mathbb{E}[Y_{0i} \mid Y_{1i} > 0]$:

$$\begin{aligned} &\mathbb{E}[Y_{1i} - Y_{0i} \mid Y_{1i} > 0] \\ &\quad + \{\mathbb{E}[Y_{0i} \mid Y_{1i} > 0] - \mathbb{E}[Y_{0i} \mid Y_{0i} > 0]\}. \end{aligned}$$

The second term is selection bias: treatment changes who ends up in the positive-outcome group, so the two groups being compared are not the same population.

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